

Impact of Media on Autonomous Buses

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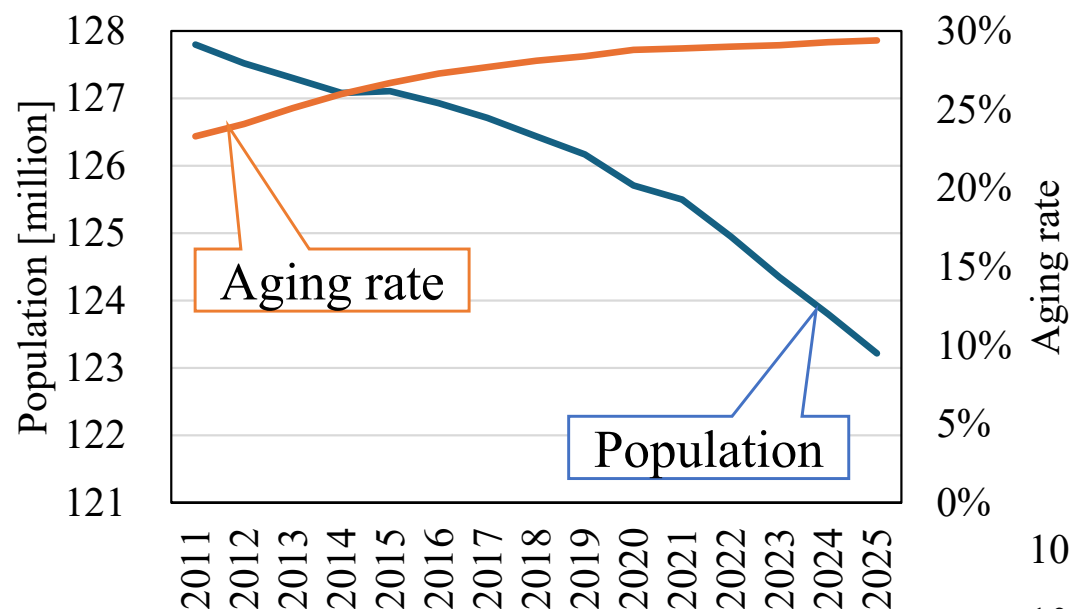
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Japanese circumstances (1)

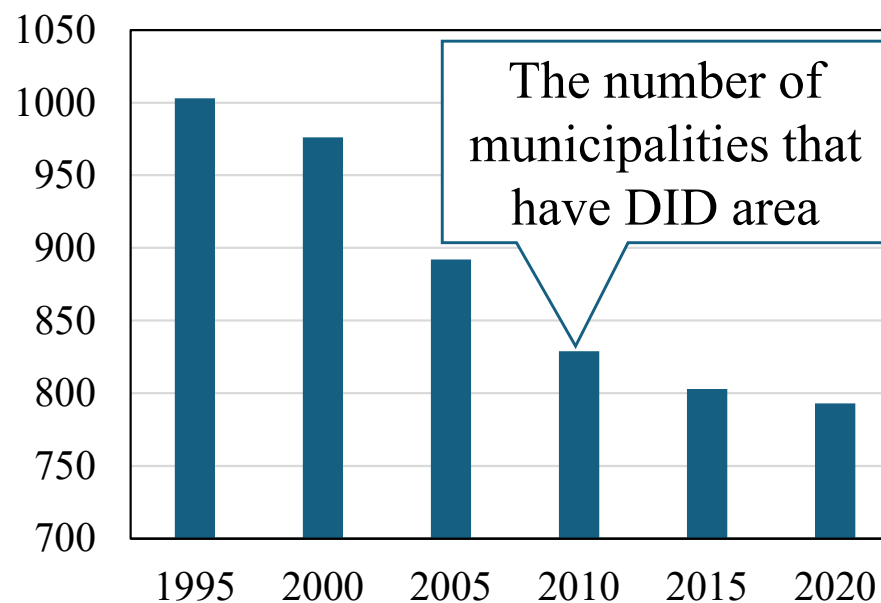


Current challenges in Japan:

- Severe population decline
- Rapidly rising aging rate
- Disappearing DID areas

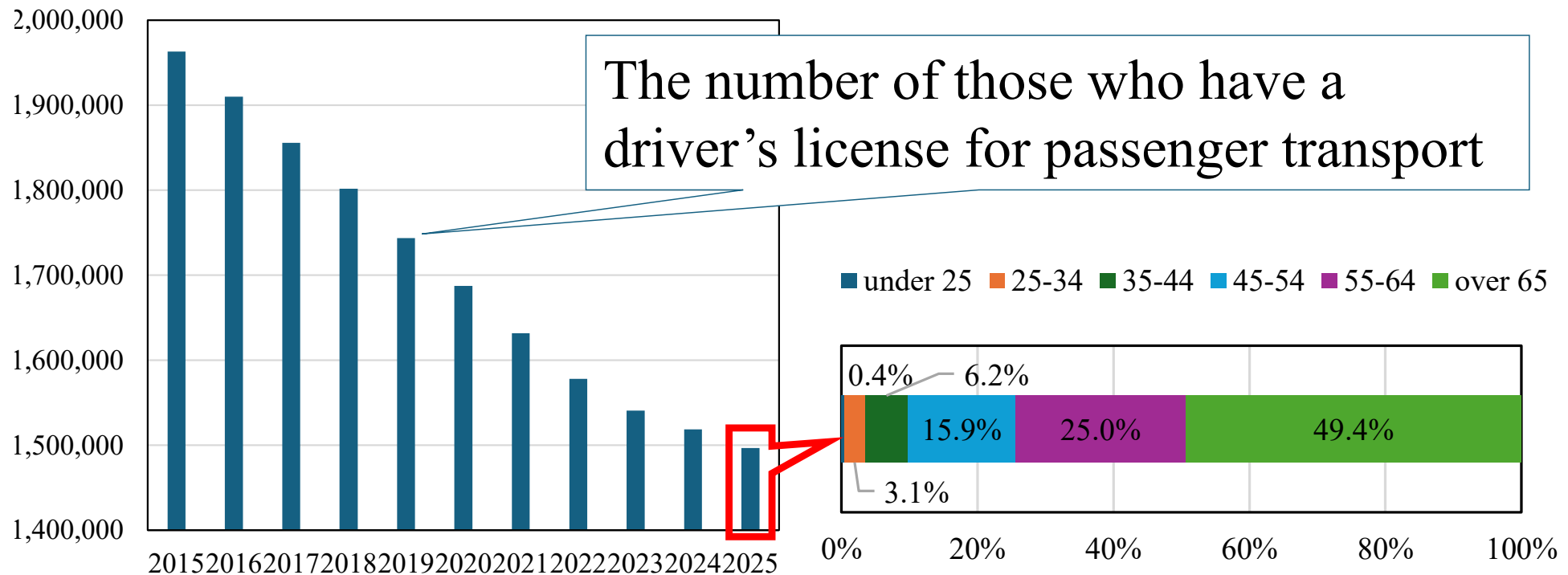
To maintain urban sustainability:

- Making city centers more compact
- Increasing public transit density are the urgent issues.



※DID (Densely Inhabited District) area:
population density > 4,000 and
total population > 5, 000

Japanese circumstances(2)



- Many Japanese bus companies are private companies.
- They are required to be financially independent of the government.

Establishing a bus system resilient to population decline and aging is necessary in Japan.

⇒ Autonomous buses are one of the strong solutions.

Autonomous Vehicle

Current status in Japan:

- Only a very limited number of level 3 households' cars are sold
- Autonomous bus operations remain limited to some trial implementations.

However, traffic accidents during trial implementations decrease the social acceptability of autonomous driving technology because they are reported sensationally through various media.

Level 0: Non-automated vehicles

Level 1: Vehicles that can assist drivers

Level 2: Vehicles that allow drivers to “hands-off” in specific areas

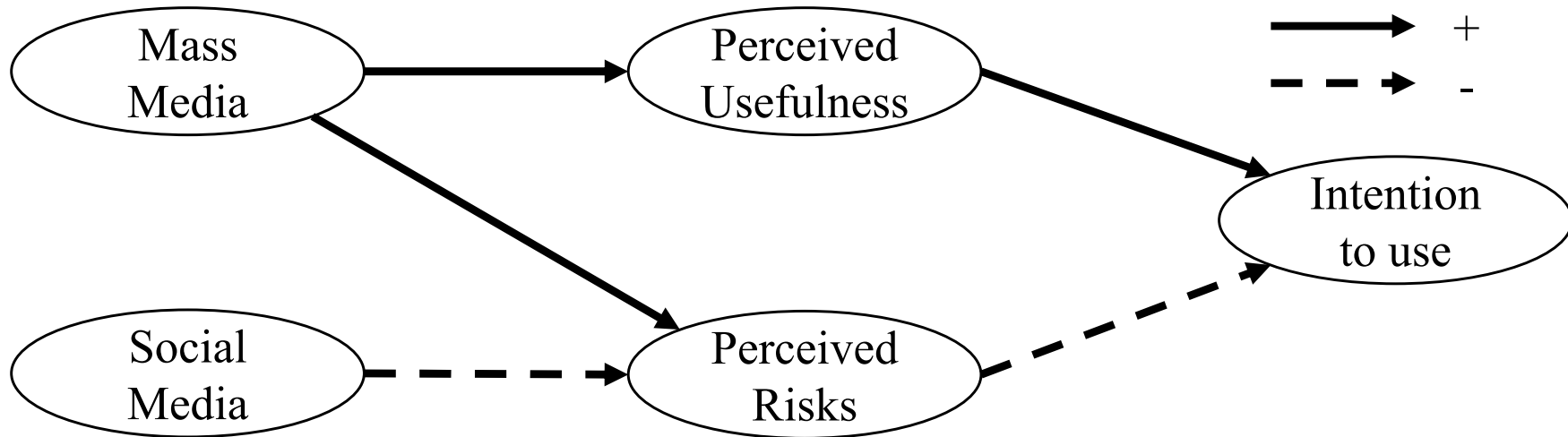
Level 3: Vehicles that allow drivers to “eyes-off” in specific areas

Level 4: Vehicles that allow drivers to “brain-off” in specific areas

Level 5: Vehicles that can drive automatically, “driver-off”

Impact of media

Zhu et al. (2020) revealed the relationship below.



Media preferences significantly affect the perceived usefulness and risks of autonomous vehicles, which affect the intention to use.

The remaining questions:

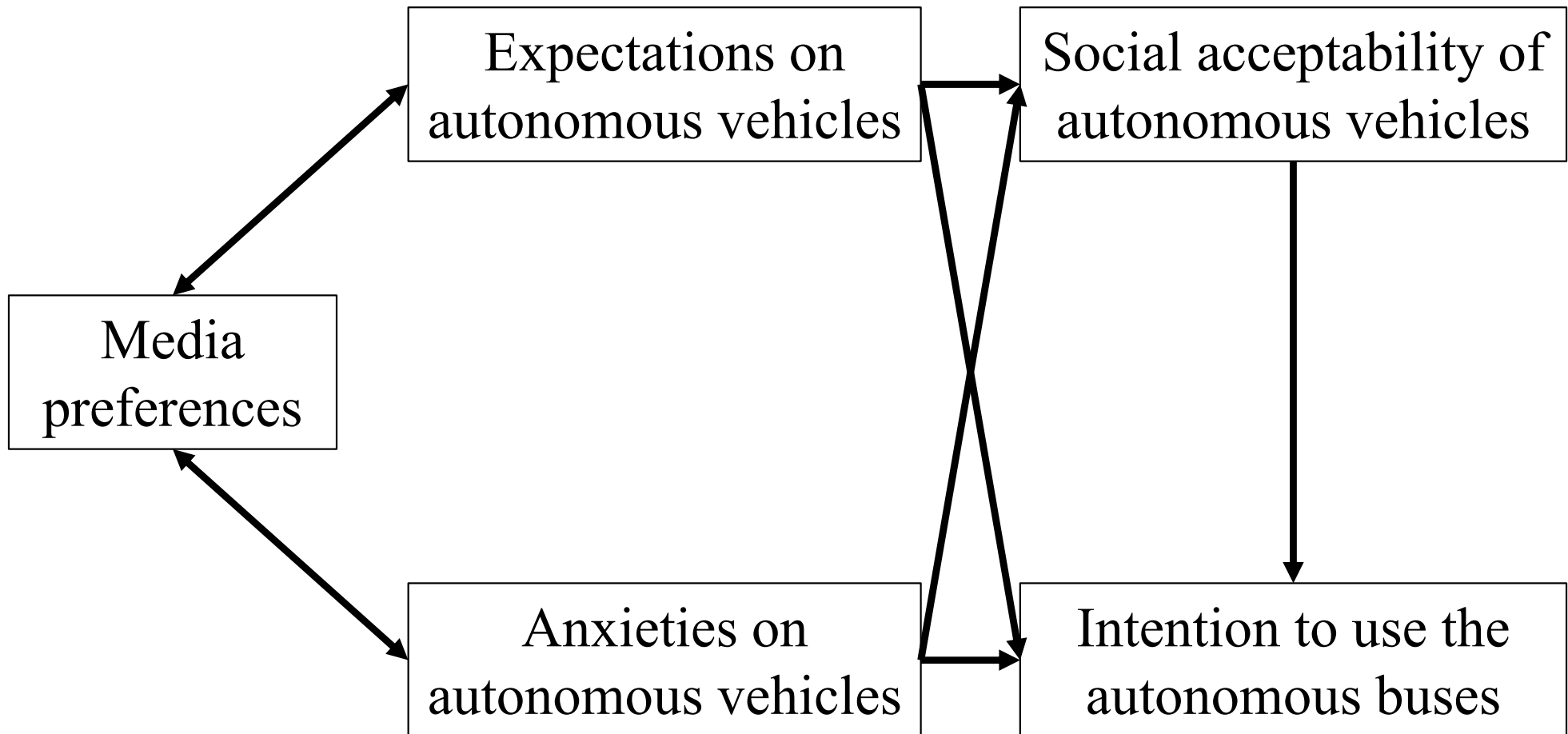
- Can individual characteristics affect this result?
- What kinds of mass media or social media are significant?
- What kinds of expectations or anxieties are significant?

Research Objective

Investigating the relationship below using a questionnaire.

(Also considered the individual characteristics.)

⇒ Targeting 724 Japanese drivers (at least once a week)



The relationship between media preference and expectations or anxieties

Variables

Expectations / Anxieties (Binary variable)

- Respondents chose up to three expectations(anxieties) they have.
(Level 3 and level 5 are considered.)

Media preferences (Binary variable)

- Respondents chose their most frequently used medium.

Individual characteristics

- Age
- Awareness of driving risk
 - “I feel that driving cars causes risks such as accidents”
(1: Strongly disagree – 5: Strongly agree)
- Preference for new products (Potential of early adopters)
 - “ I always gather information to catch up with new products and services” (1: Strongly disagree – 5: Strongly agree)

Media selection

The relationship between media preference and non-expectations or non-anxieties by χ^2 -test.

Most preferred media	χ^2			
	Expectations		Anxieties	
	Level 3	Level 5	Level 3	Level 5
TV	14.6	18.7	20.2	25.7
Radio	1.15	1.01	1.47	1.37
Newspaper	3.72	8.76	2.03	3.95
Printed ad	2.43	2.32	1.70	1.64
Magazine	3.66	3.43	2.10	1.98
Online news	19.7	18.3	13.7	19.5
Online ad	0.800	2.09	0.159	1.08
SNS	10.7	12.2	4.39	5.69

□ : $p < 0.05$, ■ : $p < 0.01$

All significant cells indicate that those who prefer each medium have an expectation/anxiety.

Summary of impact of media on expectations

Expectations	Level	Age	DR※	EA※	TV			ON※			SNS			NP※		
Effective use of travel time	3															
	5															
Driving on the disliked occasions	3															
	5															
Preventing getting lost	3															
	5															
Reducing fatigue while driving	3															
	5															
Preventing traffic accidents	3															
	5															
Easing traffic congestion	3															
	5															
Resolving the shortage of bus or truck drivers	3															
	5															
Allows people without licenses to use cars	3															
	5															
Reducing environmental impact	3															
	5															

※DR = **D**riving **R**isk, EA = **E**arly **A**dopter, ON = **O**nline **N**ews, NP = **N**ews **P**aper
 □ : p < 0.05, ■ : p < 0.01 (Positive relationship), □ : p < 0.05, ■ : p < 0.01 (Negative relationship)

Summary of impact of media on expectations

Expectations	Level	Age	DR※	EA※	TV	ON※	SNS	NP※
Effective use of travel time	3				↑Relationship between media preference and individual characteristics (Left: Age, Middle: DR, Right: EA) ←Relationship between expectations and individual characteristics			
	5							
Driving on the disliked occasions	3							
	5							
Preventing getting lost	3							
	5							
Reducing fatigue while driving	3							
	5							
Preventing traffic accidents	3							
	5							
Easing traffic congestion	3							
	5							
Resolving the shortage of bus or truck drivers	3							
	5							
Allows people without licenses to use cars	3							
	5							
Reducing environmental impact	3							
	5							

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Preventing getting lost	3							
	5							
Reducing fatigue while driving	3							
	5							
Preventing traffic accidents	3							
	5							
Easing traffic congestion	3							
	5							
Resolving the shortage of bus or truck drivers	3							
	5							
Allows people without licenses to use cars	3							
	5							
Reducing environmental impact	3							
	5							

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Driving on the disliked occasions	3							
	5							
Preventing getting lost	3							
	5							
Reducing fatigue while driving	3							
	5							
Preventing traffic accidents	3							
	5							
Easing traffic congestion	3							
	5							
Resolving the shortage of bus or truck drivers	3							
	5							
Allows people without licenses to use cars	3							
	5							
Reducing environmental impact	3							
	5							

※DR = Driving Risk, EA = Early Adopter, ON = One News, NP = News Paper

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Effective use of travel time	3							
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Preventing getting lost	3							
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Reducing fatigue while driving	3							
	5							
Preventing traffic accidents	3							
	5							
Easing traffic congestion	3							
	5							
Resolving the shortage of bus or truck drivers	3							
	5							
Allows people without licenses to use cars	3							
	5							
Reducing environmental impact	3							
	5							

※DR = Driving Risk, EA = Early Adopter, ON = Online News, NP = News Paper
 □ : p < 0.05, ■ : p < 0.01 (Positive relationship), □ : p < 0.05, ■ : p < 0.01 (Negative relationship)

Summary of impact of media on anxieties

Anxieties	Level	Age	DR	EA	TV	ON	SNS	NP
Worrying about whether I can use the autonomous driving	3							
	5							
Increasing car prices	3							
	5							
Increasing maintenance costs	3							
	5							
Loss of enjoyment in driving	3							
	5							
Decline in driving skills among autonomous vehicle users	3							
	5							
Increasing traffic accidents	3							
	5							
Mixing of regular and autonomous cars	3							
	5							
Responsibility for accidents while using autonomous driving	3							
	5							
System malfunction	3							
	5							
Technical failure due to hacking, etc.	3							
	5							

Summary of impact of media on anxieties

Anxieties	Level	Age	DR	EA	TV	ON	SNS	NP
Worrying about whether I can use the autonomous driving	3							
	5							
Increasing car prices	3							
	5							
Increasing maintenance costs	3							
	5							
Loss of enjoyment in driving	3							
	5							
Decline in driving skills among autonomous vehicle users	3							
	5							
Increasing traffic accidents	3							
	5							
Mixing of regular and autonomous cars	3							
	5							
Responsibility for accidents while using autonomous driving	3							
	5							
System malfunction	3							
	5							
Technical failure due to hacking, etc.	3							
	5							

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Increasing car prices	3							
	5							
Increasing maintenance costs	3							
	5							
Loss of enjoyment in driving	3							
	5							
Decline in driving skills among autonomous vehicle users	3							
	5							
Increasing traffic accidents	3							
	5							
Mixing of regular and autonomous cars	3							
	5							
Responsibility for accidents while using autonomous driving	3							
	5							
System malfunction	3							
	5							
Technical failure due to hacking, etc.	3							
	5							

The relationship between expectations or
anxieties and social acceptability or
intention to use

Variables

- Social acceptability of autonomous vehicles
“Regardless of whether they are private cars or public transit, how do you perceive the widespread adoption of driverless autonomous vehicles from a social perspective?”
(1: Bad – 5: Good)
- Intention to use autonomous buses
”If autonomous driving public transit, such as buses or taxis, is implemented, would you want to use it?” (1: No – 5: Yes)

Individual characteristics

- Preference for new products (Potential of early adopters)
“New products and services appeal to me”
(1: Strongly disagree – 5: Strongly agree)
※It is different from the previous analysis.

Result (Multiple regression analysis: Characteristics)

↓Independent variables	Dependent variables→			
	Acceptability		Intention	
	(R ² =0.140)		(R ² =0.227)	
Const.	3.809	**	1.223	**
Social acceptability of autonomous driving system			0.396	**
I feel that driving cars causes risks such as accidents	0.061		-0.027	
New products and services appeal to me	0.028		0.137	**
Age	0.002		-0.003	
	λ	λ		
*: p < .05, **: p < .01				

- Social acceptability is not affected by the individual characteristics.
- In contrast, intention to use is significantly driven by a preference for new things (early adopters).
⇒ Those who like new products tend to have the intention to use.
- Higher social acceptability strongly promotes intention to use.

Result (Multiple regression analysis: Expectations)

↓Independent variables	Dependent variables→ Acceptability Intention		
	(R ² =0.140)		(R ² =0.227)
	λ	λ	
Effective use of travel time(reading, eating, sleeping, etc.)	0.354	**	0.056
Driving on the disliked occasions	0.056		0.157 *
Preventing getting lost	0.085		0.048
Reducing fatigue while driving	0.088		0.051
Preventing traffic accidents	0.183	*	0.118
Easing traffic congestion	0.115		0.108
Resolving the shortage of bus or truck drivers	0.225	**	0.014
Allows people without licenses to use cars	0.298	**	0.160
Reducing environmental impact	-0.016		0.048
	λ	λ	
*: p < .05, **: p < .01			

- Various expectations can foster social acceptability, but only one expectation can foster intention to use directly.
- However, social acceptability can mediate various expectations and intention to use.

Summary of impact of media on expectations

Expectations	Level	Age	DR※	EA※	TV	ON※	SNS	NP※
Effective use of travel time	3							
	5							
Driving on the disliked occasions	3							
	5							
Preventing getting lost	3							
	5							
Reducing fatigue while driving	3							
	5							
Preventing traffic accidents	3							
	5							
Easing traffic congestion	3							
	5							
Resolving the shortage of bus or truck drivers	3							
	5							
Allows people without licenses to use cars	3							
	5							
Reducing environmental impact	3							
	5							

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Summary of impact of media on expectations

Expectations	Level	Age	DR※	EA※	TV	ON※	SNS	NP※
Effective use of travel time	3							
	5							
Driving on the disliked occasions	3							
	5							
Preventing getting lost	3							
	5							
Reducing fatigue while driving	3							
	5							
Preventing traffic accidents	3							
	5							
Easing traffic congestion	3							
	5							
Resolving the shortage of bus or truck drivers	3							
	5							
Allows people without licenses to use cars	3							
	5							
Reducing environmental impact	3							
	5							

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 □ : p < 0.05, ■ : p < 0.01 (Positive relationship), □ : p < 0.05, ■ : p < 0.01 (Negative relationship)

Result (Multiple regression analysis: Anxieties)

↓Independent variables	Dependent variables→ Acceptability Intention	
	(R ² =0.140)	(R ² =0.227)
	λ	λ
Worrying about whether I can use the autonomous driving systems	-0.084	0.033
Increasing car prices	0.051	0.085
Increasing maintenance costs	-0.135	0.022
Loss of enjoyment in driving	-0.009	-0.012
Decline in driving skills among autonomous vehicle users	-0.085	-0.236 **
Increasing traffic accidents	-0.354 **	-0.153
Mixing of regular and autonomous cars	-0.023	0.032
Responsibility for accidents while using autonomous driving systems	-0.037	-0.004
System malfunction	-0.051	-0.001
Technical failure due to hacking, etc.	0.026	0.061
*: p < .05, **: p < .01		

- Same as existing literature, traffic accidents are the key factor for social acceptability.
- Regarding the impact related to traffic accidents, anxiety has a stronger effect than expectation.

Summary of impact of media on anxieties

Anxieties	Level	Age	DR	EA	TV	ON	SNS	NP
Worrying about whether I can use the autonomous driving	3							
	5							
Increasing car prices	3							
	5							
Increasing maintenance costs	3							
	5							
Loss of enjoyment in driving	3							
	5							
Decline in driving skills among autonomous vehicle users	3							
	5							
Increasing traffic accidents	3							
	5							
Mixing of regular and autonomous cars	3							
	5							
Responsibility for accidents while using autonomous driving	3							
	5							
System malfunction	3							
	5							
Technical failure due to hacking, etc.	3							
	5							

Key findings

- Online news has a relatively stronger impact on both expectations and anxieties.
- The expectations and anxiety related to social acceptability can be affected by online news.
- Online news can affect intention to use autonomous buses through social acceptability.
- TV, newspapers, and SNS have relatively weaker impacts, so they need to be used more efficiently.
- SNS may lead to misunderstandings.
- Regarding traffic accidents, anxiety has a stronger impact.

Key findings

- TV tends to have stronger impacts on anxieties than expectations.
- Internet-based media tend to cause anxieties that tend to decrease social acceptability and intention to use.

Limitations

- These results are based on Japanese drivers, so they cannot be applied directly to other countries.
- Autonomous driving technology has not been implemented, so social acceptability and intention to use are based on the image of each respondent.
- These relationships are examined one by one.

Thank you for listening.

Appendix: Media selection

Most preferred media		Level 3				Level 5			
		No expectation		No anxiety		No expectation		No anxiety	
		Yes	No	Yes	No	Yes	No	Yes	No
TV	Yes	54	330	34	350	49	335	30	354
	No	86	254	70	270	86	254	71	269
Radio	Yes	2	18	1	19	2	18	1	19
	No	138	566	103	601	133	571	100	604
News paper	Yes	10	76	8	78	6	80	6	80
	No	130	508	96	542	129	509	95	543
Printed ad	Yes	0	10	0	10	0	10	0	10
	No	140	574	104	610	135	579	101	613
Magazine	Yes	1	23	1	23	1	23	1	23
	No	139	561	103	597	134	566	100	600
Online news	Yes	8	129	6	131	8	129	3	134
	No	132	455	98	489	127	460	98	489
Online ad	Yes	2	16	2	16	1	17	1	17
	No	138	568	102	604	134	572	100	606
SNS	Yes	3	65	4	64	2	66	3	65
	No	137	519	100	556	133	523	98	558
Other online site	Yes	2	22	3	21	4	20	2	22
	No	138	562	101	599	131	569	99	601

 : large ($p < 0.01$),
 : small ($p < 0.01$),
 : large ($p < 0.05$),
 : small ($p < 0.05$)

Appendix: Expectations

	Level 3	Level 5
Effective use of travel time (reading, eating, sleeping, etc.)	78	180
Driving on the disliked occasions	231	240
Preventing getting lost	159	195
Reducing fatigue while driving	306	286
Preventing traffic accidents	240	244
Easing traffic congestion	101	159
Resolving the shortage of bus or truck drivers	95	176
Allows people without licenses to use cars	45	131
Reducing environmental impact	35	60

Appendix: Anxieties

	Level 3	Level 5
Worrying about whether I can use the autonomous driving systems	177	328
Increasing car prices	309	328
Increasing maintenance costs	237	260
Loss of enjoyment in driving	116	249
Decline in driving skills among autonomous vehicle users	146	194
Increasing traffic accidents	102	175
Mixing of regular and autonomous cars	171	159
Responsibility for accidents while using autonomous driving systems	208	137
System malfunction	290	128
Technical failure due to hacking, etc.	99	103